DAYANANDA SAGAR UNIVERSITY

SCHOOL OF ENGINEERING

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DEPARTMENT OF COMPUTER SCIENCE & TECHNOLOGY

Assignment-1: Case Study/ Field Work

"Identify a real-world problem and propose a product concept by applying principles of product planning, opportunity identification, and customer need analysis."

Project Title: EazePrint: AI-Powered Hyperlocal Cloud Printing Platform

Team Number: 21

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1. Product Planning

Product Vision

To create a cloud-based, hyperlocal printing network that enables users to send print jobs remotely and collect them instantly from nearby print shops, eliminating waiting time and manual processes.

Product Objectives

- To eliminate waiting time at print shops
- To digitize and simplify the traditional printing process
- To provide a fast, reliable, and user-friendly printing solution
- To connect customers and local print shops through a unified platform
- To improve efficiency and productivity of print shop owners
- To build a scalable printing infrastructure across multiple cities

Target Market & Intended Users

Primary Users (Customer Side):

- Students (schools, colleges, universities)
- Professionals (law firms, offices, freelancers)

Secondary Users (Service Providers):

- Print shop owners
- Small businesses requiring bulk printing

Geographic Market:

- Initially urban and semi-urban areas (e.g., Bangalore)
- Future expansion to tier-2 and tier-3 cities across India

Scope of the Product

Customer Side Scope

- Upload documents digitally from anywhere
- Customize print options (pages, color, copies, size)
- Discover nearby print shops
- Send print requests remotely
- Visit shop only for quick pickup

Shopkeeper Side Scope

- Receive print jobs from customers through the platform

- Manage incoming orders digitally
- Automate printing using connected systems
- Handle multiple customers efficiently
- Increase business through online visibility

Key Features of the Product

Customer Side Features

- Easy document upload (PDF, Word, images)
- Print customization options
- Location-based shop discovery
- Real-time order tracking
- Estimated print time before arrival
- Transparent pricing

Shopkeeper Side Features

- Dashboard to receive and manage print jobs
- Accept/reject orders
- View and manage print queue
- Track workload and order status

EazePrint Agent (Core Feature – Shop Side)

- Installed on shop computers
- Automatically receives print jobs from the cloud
- Downloads files securely
- Sends documents directly to the printer
- Manages print queue without manual intervention

Additional Features

- Secure document handling
- Fast cloud-based job transfer
- Scalable system for multiple users and shops

2. Opportunity Identification

Existing Solutions in the Market

Currently, printing needs are fulfilled through a mix of traditional and digital solutions:

Offline Print Shops

- Local shops where users physically visit to print documents
- No digital integration or pre-booking system
- Highly dependent on manual communication

Email / WhatsApp-Based Printing

- Users send files via email or messaging apps
- Shopkeepers manually download, manage, and print files
- No structured queue or tracking system

Online Printing Services

Examples include platforms like Printo and Vistaprint

- Offer bulk printing and delivery services
- Focus more on business orders rather than instant local needs
- Longer delivery time, not suitable for urgent prints

Gaps & Limitations in Current Solutions

Despite multiple options, several critical problems still exist:

Lack of Convenience

- Users must physically visit shops or coordinate manually
- No seamless "upload → print → pickup" workflow

No Real-Time Availability

- Users don't know:
    - Which shop is open
    - Printer status
    - Queue length

Inefficient Job Handling

- Shopkeepers manage jobs manually
- High chance of:
    - Misplaced files
    - Wrong prints

    - Delays

No Standardization or Tracking

- No order tracking system
- No payment integration
- No digital receipts or history

No Hyperlocal Optimization

Existing platforms are not optimized for:

    - Nearby shop discovery
    - Fast turnaround (within minutes)

Missed Opportunities for Small Businesses

Local print shops lack:

    - Digital presence
    - Customer reach
    - Automated workflow tools

Unmet Needs Identified

From the above gaps, the following unmet needs are clear:

A platform for **instant document upload and printing**
Hyperlocal shop discovery (nearest available printer)
Real-time job tracking and queue management
Automated workflow for shopkeepers
Integrated payment system
Faster turnaround (within minutes, not days)

Why This is a Strong Product Opportunity

High Demand (Daily Use Case)

- Students, professionals, and businesses frequently need printing
- Especially in India:

    - Colleges
    - Offices
    - Government services

Urgency Factor

- Printing is often **last-minute**
- Current systems fail in urgent scenarios

Digital Transformation Gap

- Many industries are digitized, but **local printing is still manual**
- Huge scope for modernization

Win-Win Ecosystem

- **Users** → convenience + speed
- **Shopkeepers** → more orders + automation
- Creates a scalable two-sided platform

Scalability

- Can expand to:
    - Multiple cities
    - Franchise model
    - Additional services (scanning, lamination, etc.)

Final Justification

The problem represents a **highly viable product opportunity** because it lies at the intersection of:

Frequent demand
Clear inefficiencies
Lack of strong existing solutions
High scalability potential

EazePrint addresses these gaps by creating a smart, cloud-based, hyperlocal printing network that enables seamless, real-time, and efficient printing for users while empowering local print shops with digital tools.

3. Identifying Customer Needs

Primary Customer Needs

Fast and time-saving printing without waiting in queues

Easy and convenient document upload

Access to nearby print shops

Reliable and accurate printing

Quick turnaround time

Secondary Customer Needs

Affordable pricing

Order tracking and status updates

Secure handling of documents

Availability of multiple print options (color, size, copies)

Support for bulk printing

Customer Persona

Name: Rahul Sharma
Age: 20
Occupation: College Student
Location: Bangalore

Background:
Rahul is a college student who frequently needs to print assignments, lab records, and notes. He usually visits nearby print shops before classes.

Problems Faced:

Long waiting time at print shops

Difficulty in sending files through WhatsApp or other methods

Delays during urgent printing needs

Goals:

To get printouts quickly without waiting

To upload documents easily from mobile or laptop

To ensure prints are ready before reaching the shop

Translation of Customer Needs into Product Requirements

Customer Need	Product Requirement
No waiting time	Documents should be printed before customer arrival
Easy file submission	Provide web/app-based document upload system
Nearby access	Enable location-based print shop discovery
Fast service	Use cloud-based job transfer to shops
Reliable printing	Implement automated printing system
Cost transparency	Show pricing before order confirmation
Order updates	Provide real-time order tracking

4. Conclusion

EazePrint effectively addresses the major challenges in traditional printing systems by introducing a **smart, cloud-based, and hyperlocal solution**. It eliminates the need for physical visits, manual coordination, and uncertainty by enabling users to upload documents, select nearby print shops, and receive prints quickly and efficiently.

The platform creates significant value for **customers** by providing:

- Convenience and time savings
- Real-time tracking and transparency
- Faster, on-demand printing services

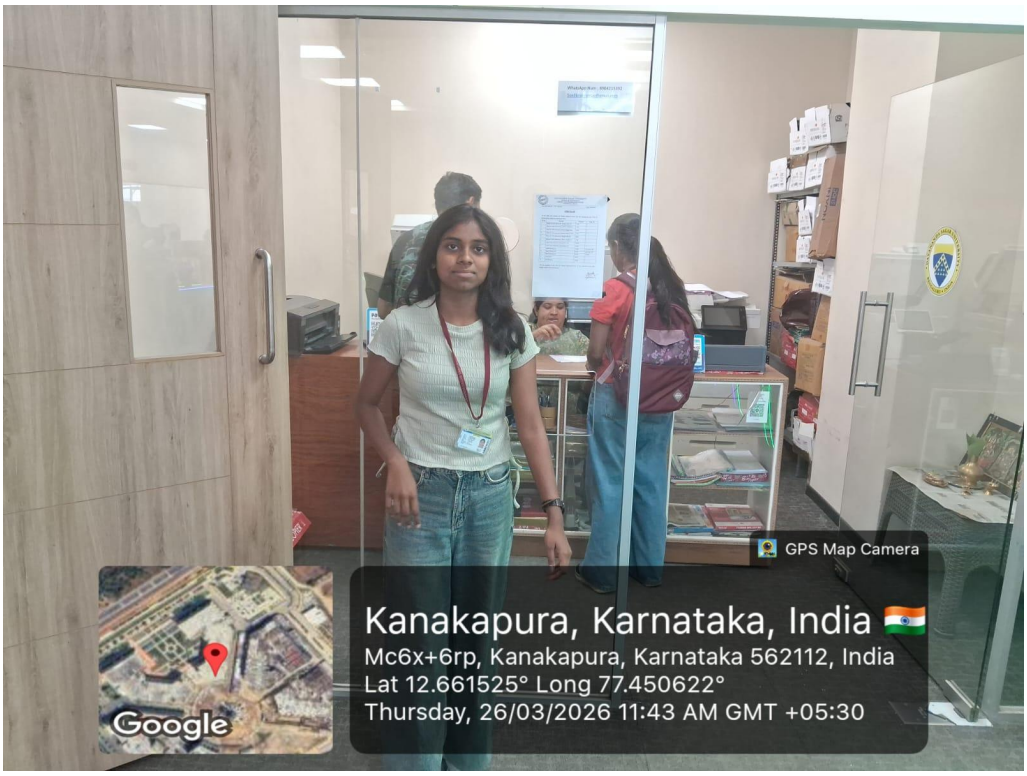
At the same time, it empowers **local print shop owners** by:

- Digitizing their workflow
- Increasing order volume through better visibility
- Reducing manual errors and inefficiencies

By bridging the gap between users and local vendors through technology, EazePrint builds a **scalable and sustainable ecosystem**. It not only improves user experience but also drives digital transformation for small businesses.

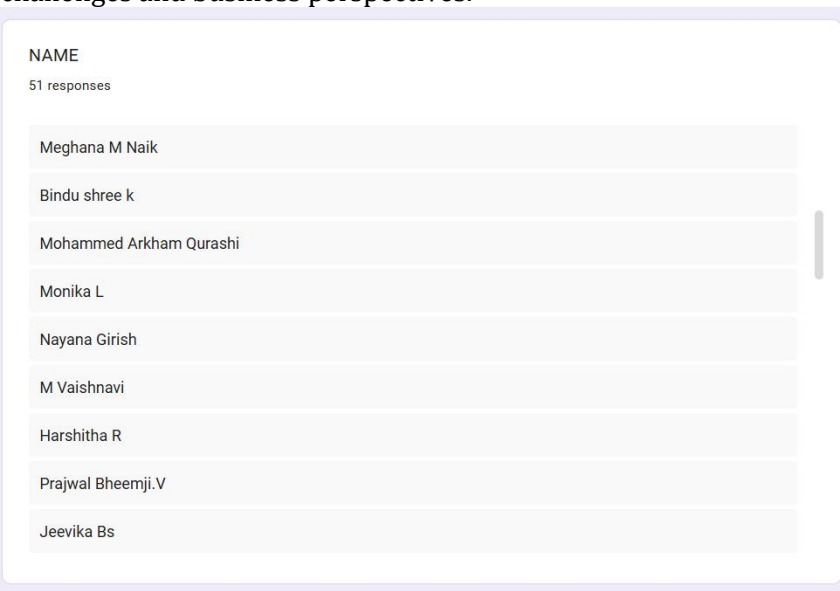
Overall, EazePrint transforms printing into a **seamless, accessible, and efficient service**, making it a valuable and impactful product opportunity.

PhotoGraphs of the Fieldwork (Geotag)



As part of our research and validation process, we conducted both user and vendor surveys.

- We collected **51 responses from users** as part of our literature and requirement survey.
- Additionally, we surveyed **10 Xerox (printing) shops** to understand operational challenges and business perspectives.



NAME

51 responses

Meghana M Naik

Bindu shree k

Mohammed Arkham Qurashi

Monika L

Nayana Girish

M Vaishnavi

Harshitha R

Prajwal Bheemji.V

Jeevika Bs

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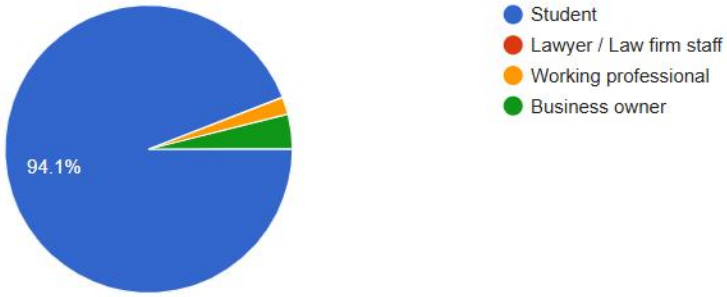
Copy chart

What best describes you?

51 responses



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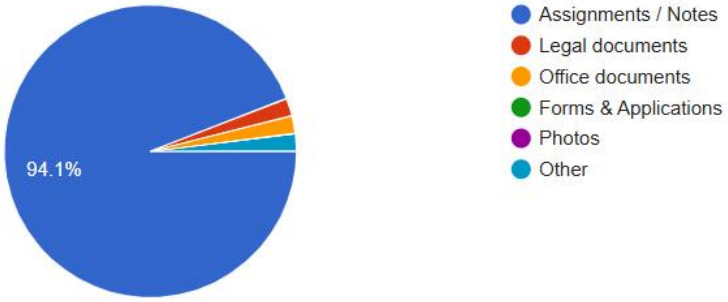
What type of documents do you usually print?

51 responses



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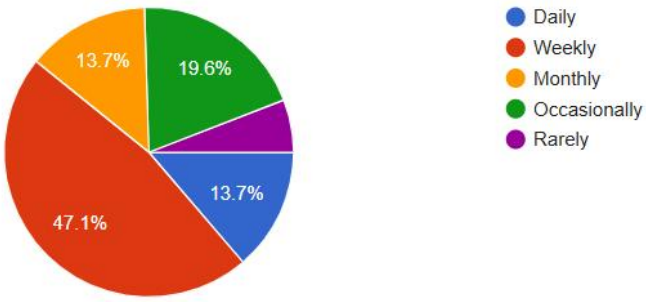
How often do you need printouts?

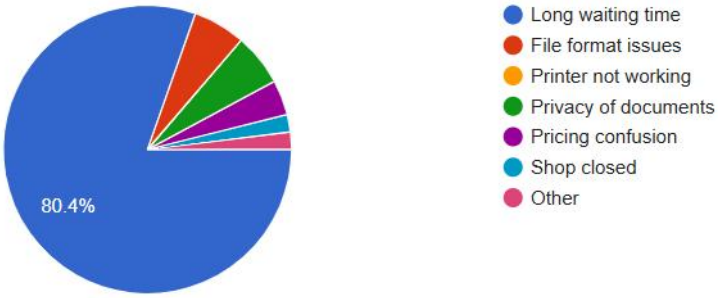
51 responses



What worries you before going to a print shop?

Copy chart

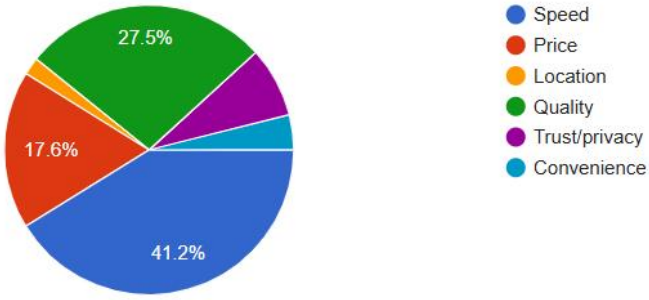
51 responses



When choosing a print shop, what matters most?

Copy chart

51 responses



DO YOU WANT TO SIT ANYWHERE AND SEND THE PRINTS TO THE SHOP AND LATER COLLECT IT

Copy chart

51 responses